

SoIM
School of Innovation and Management
Innovate. Include. Inspire.

TECHNICAL ANALYSIS WITH TRADING STRATEGIES

Dr.RAJKUMAR RAYABARAM
MBA,MPhil,PhD,UGC-NET-JRF,AP-SET,MISTE
Professor in Finance & Analytics

What is investment

- Investment involves making of a sacrifice in the present with the hope of deriving future benefits
- Investment has many meanings however; investment can be interpreted broadly from three angles -
  - economic angle
  - Layman angle
  - Financial angle

Economic angle

- Economic investment includes the commitment of the fund for net addition to the capital stock of the economy.
- The net additions to the capital stock means an increase in building equipments or inventories over the amount of equivalent goods that existed, say, one year ago at the same time
- Example: setting up a business

Layman Angle

- The layman uses of the term investment as any commitment of funds for a future benefit not necessarily in terms of return
- For example a commitment of money to buy a new car is certainly an investment from an individual point of view

Financial angle

- Financial investment is the commitment of funds for a future return
- Hence the purchase of shares, debentures post office savings certificates and insurance policies all are financial investments
- Financial investments are commitments of funds to derive income in form of interest, dividend premium, pension benefits or appreciation in the value of initial investment
- In the present context of portfolio management, the investment is considered to be financial investment, which imply employment of funds with the objective of realizing additional income or growth in value of investment at a future date

Investment vs. Speculation

	INVESTMENT	SPECULATION
TIME HORIZON	Long-term, saving for future	Generally a short timeline of less than one year
LEVEL OF RISK	Moderate	High
INVESTOR ATTITUDE	Cautious and conservative	Aggressive
DECISION CRITERIA	Based on fundamental and basic factors	Based on technical charts, market psychology and individual opinion
EXAMPLES	Stock market, bonds, mutual funds	Options, foreign currencies, cryptocurrencies

Source: U.S. Global Investors

Real vs Financial assets



Real assets are value-driven physical assets that a company owns



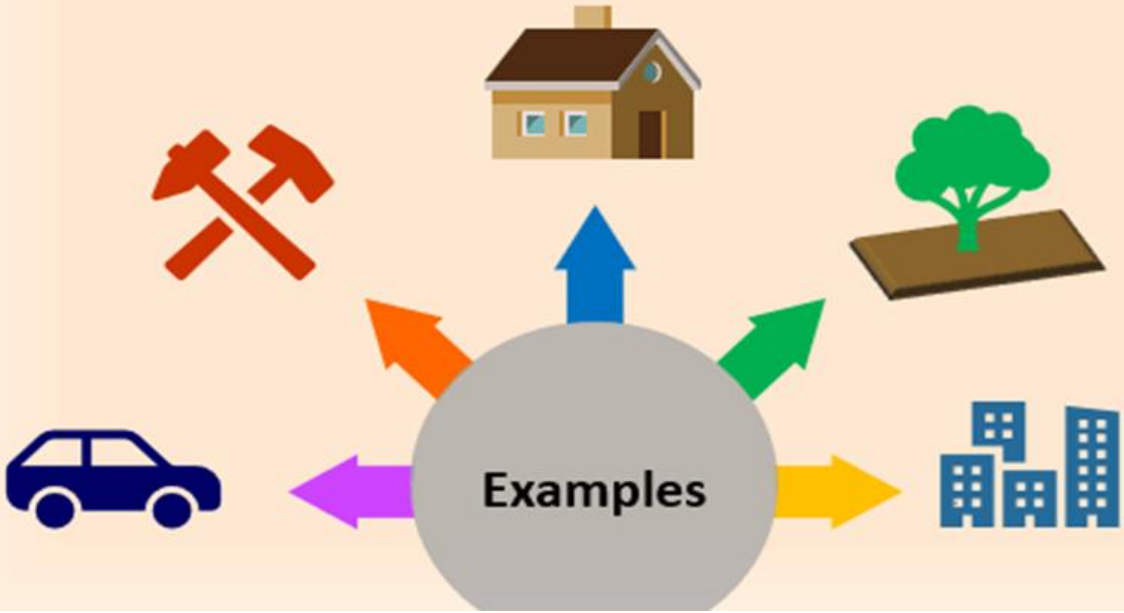
They include land, buildings, motor car or commodities



The unique feature of it is that they have intrinsic value by itself and don’t rely on exchanges to have value

compiled by Dr.RAJKUMAR R

Real Assets



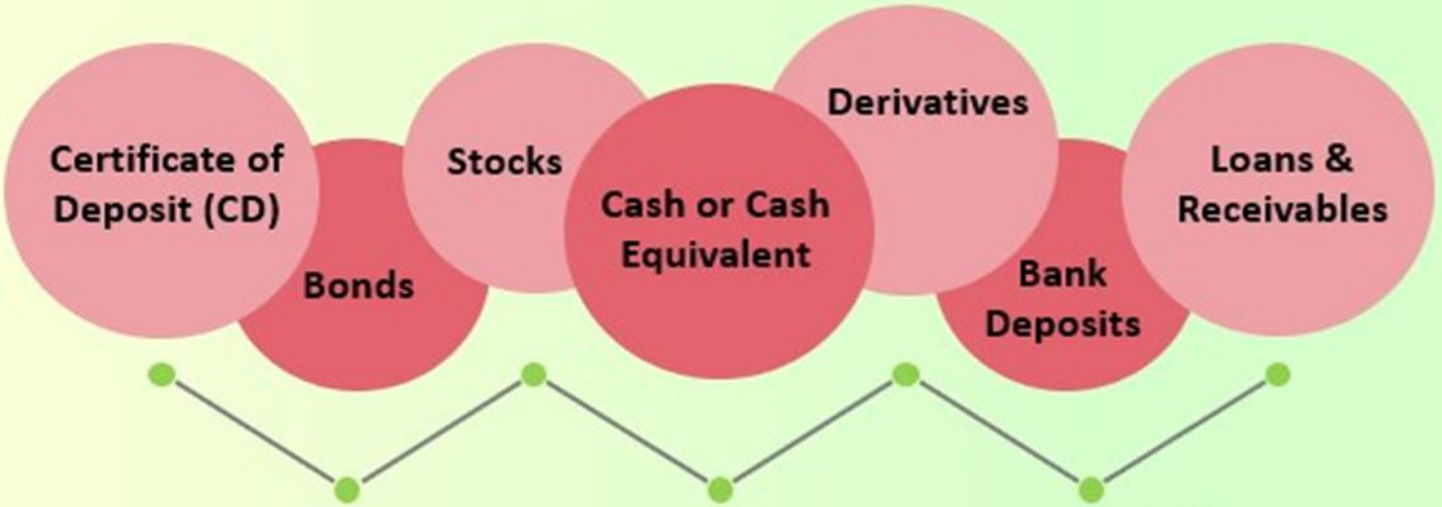


Financial Assets

- **Financial Assets** are highly liquid assets that are either in cash or can be fast converted to cash
- They include investments such as stocks and bonds
- The major feature of financial assets is that it has some economic value that is easily realized However, by itself, it has lesser intrinsic value

Financial Assets

Types



Investment decision process



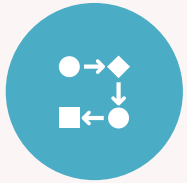
STEP-1. SET INVESTMENT POLICY (INVESTABLE FUNDS AND OBJECTIVES AND KNOWLEDGE)



STEP-2 ANALYSIS (MARKET INDUSTRY AND COMPANY)



STEP-3 VALUATION (INTRINSIC VALUE FUTURE VALUE)



STEP-4 PORTFOLIO CONSTRUCTION (DIVERSIFICATION, SELECTION AND ALLOCATION)



STEP-5 EVALUATE PERFORMANCE OF PORTFOLIO (APPRAISAL AND REVISION)

STEP-1. Set investment policy

- Investment policy involves determining
  - the investor's objective
  - Investible fund
  - Knowledge (risk-return)

Step-2 analysis

- Market analysis includes looking at market indicators like BSE SESNSEX, NSE-50
- Industry analysis includes growth prospects of industry say IT industry
- Company analysis includes company’s earnings,profitability,capital structure etc

Step-3 valuation

- **Intrinsic value of share** is measured through book value of share and price-earning ratio
  - Intrinsic value should always be greater than the market price adjusted with risks and inflation
- Future value can be estimated using a simple statistical technique like trend analysis

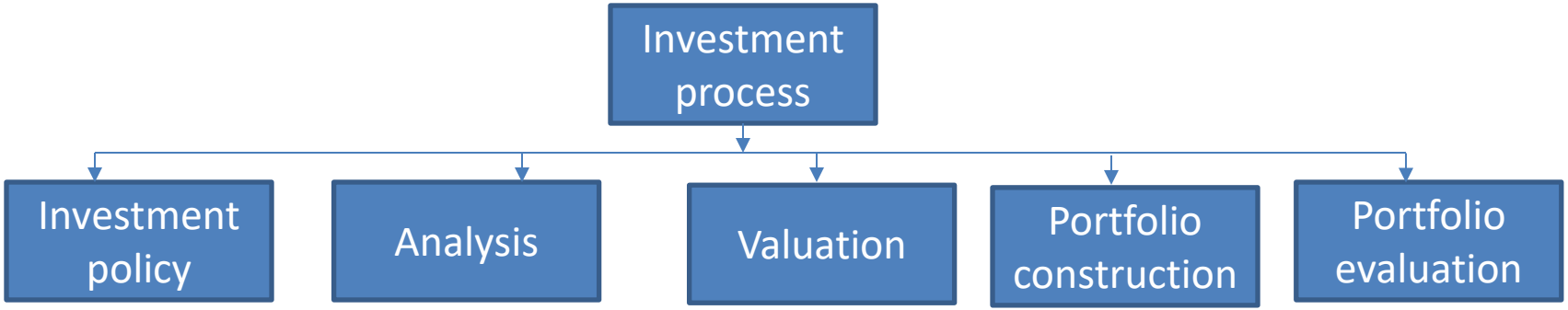
Step-4 portfolio construction

- Diversification- reduction of risk in the form of loss of capital and income

  Types:
  - Debt-equity diversification
  - Industry diversification
  - Company diversification
- Selection and allocation- securities have to be selected based on the level of diversification- industry and company analysis

Step-5 evaluate performance of portfolio

- **Appraisal** – the variability in returns of securities is measured and compared
  - The appraisal warns about the loss and steps can be taken to such losses
- **Revision**- depends on results of the appraisal
  - Low yielding securities are replaced with high-yielding securities with low risk factor


Investment process
Investment policy
Analysis
Valuation
Portfolio construction
Portfolio evaluation

Factors to be considered in investment decision

1.Return

Investors always expect a good rate of return from their investment rate of return could be defined as the total income the investor receives during the holding period stated as a percentage of the purchasing price at the beginning of the holding period



2. Safety of funds

- The selected investment avenue should be under the legal and regulatory framework
- If it is not under the legal framework
- It is difficult to represent the grievances if any approval of the law itself adds a flavor of safety

3. Tax benefits

- The investor who invests their money in shares and securities which will reduce the investor’s income tax burden

4. Capital growth

- Every investor seeks maximum capital growth and implies making risky investment with considerable investment analysis and management

  Example – equity share and the growth shared

5.Liquidity

- If a portion of the investment could be converted into cash without much loss of time. It would help the investor meet the emergencies.

6. Stability income

- It refers to constant return from an investment. Another major character feature of the investment is the stability of income.

7. Diversification

- The idea is to create a portfolio the includes multiple investments in order to reduce risk.

8.Marketability

- It can be transacted quickly
- The transaction cost is low
- The price change between two successive transactions
- The above are the objectives of every investor to increase their wealth

Concept of risk

- Risk is defined as variability in expected return (or) variability in return or volataility in return i.e. chance of the actual return being less than expected return



compiled by Dr.RAJKUMAR R

Risk consists

Systematic risk- external factors cause systematic risk to a company- the company is not able to control this risk- it affects market as a whole e.g. bull/bear market, political situations etc.

Unsystematic risk- these factors are unique, specific and related to the industry or company e.g. managerial inefficiency, technological change in production process

Systematic risk includes

1. Market risk
   - Tangible events
   - Intangible events
2. Interest rate risk
   - Bond return
   - Cost of borrowing
3. Purchasing power risk – loss of purchasing power of the currency

Unsystematic risk

- It is unique and peculiar to a firm or industry
- Unsystematic risk includes
- **Business Risk: -** The risk of doing business in a particular industry or environment is called business risk
- **Internal business risk-** fluctuations in sales, outdated product,lockouts, high fixed costs, single product firms
- **External business risk –** social regulatory factors, political risk, business cycle
- **B. Financial Risk: -**
  - associated with the capital structure of the firm
  - Variability of the income to the equity

Why Do We Invest?

Investment is necessary to support your financial needs when you do not earn money.

1. By investing a portion of your income you allow money to grow and work for you.
2. 3 parameters to assess suitability of any investment avenue are –
   i. Return potential
   ii. Safety
   iii. Liquidity
3. Various avenues where money can be invested, are broadly classified into some groups, known as 'Asset Class'. Stocks or Equity shares are most popular class of assets.



What is meant by Stock Selection?

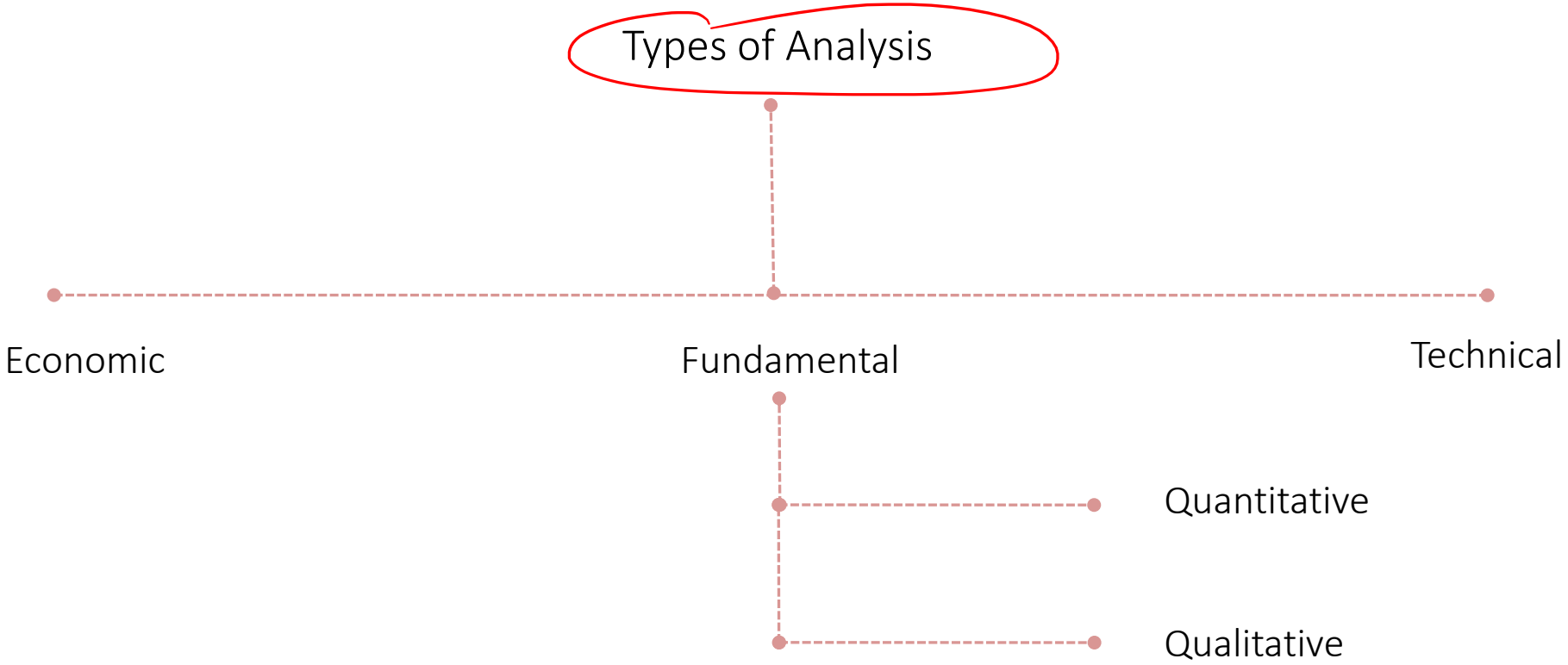
Stock selection is the selection of one or more stock (or shares) based on certain set of criteria in order to maximise the probability of meeting the trading or investment objective.

There are more than 5000 stocks available for trading or investment. None can trade or invest in all at the same time.

Hence, one needs to select a manageable number of stocks.

Fundamental and Technical Analysis are the two most preferred tools for stock selection.




Types of Analysis
Economic
Fundamental
Technical
Quantitative
Qualitative

Fundamental Analysis

Fundamental analysis is a method used to identify the true value of a stock.

1. The current price of a stock may not reflect the actual value of the stock. The stock may be overvalued or undervalued in the market.
2. Fundamental analysis helps investors to study the health of the company, and thus leading to the actual value of the stock.
3. This is done by using various qualitative and quantitative factors.
4. The main purpose of this method is to identify companies that that are fundamentally strong in order to invest in them for the long term.



Economic Analysis

1. It involves assessing or examining topics or issues from an economist's perspective.
2. This allows investors to analyse the market from the big picture to all the way down to individual stocks.
3. By examining the economic numbers one can determine the current market strength and have a better idea of what the future holds.
4. Key Economic indicators investors must incorporate while selecting stocks:

    i. Indices (e.g. Nifty, Sensex)
    ii. Gross Domestic Product (GDP)
    iii. Unemployment rate
    iv. Inflation rate
    v. Consumer Confidence
    vi. Purchase Managers' Index

Types of Fundamental Analysis

Qualitative Analysis

It takes into account information that can not be expressed in numbers.

i. It relates to the company itself.
ii. Factors – examples –

a) Management experience and performance
b) Corporate governance
c) Industry and competition etc.



Types of Fundamental Analysis

Quantitative Analysis

i. It is related to information that is shown in company's financial statements. It involves measuring simple statistical data to complex calculations.
ii. This analysis helps you to evaluate investment opportunities such as when to buy and sell securities.
iii. Factors – examples -
   a) Company's revenues
   b) Profit margins
   c) Return on equity
   d) Future growth potential
   e) Financial ratios



Using Financial Ratios for Fundamental Analysis

1. Price To Earnings Ratio

i. It is one of the most widely used financial ratio analysis.
ii. Computation - Price Per Share / Earnings Per Share.
iii. As a thumb rule, a low P/E ratio is preferred while buying a stock.

2. Price To Book Value

i. Computation - Current price of the stock / Book value per share.
ii. A lower P/BV ratio could mean that the stock is undervalued.
iii. However, the definition of lower varies from sector to sector.

Financial Ratios

3. Return On Equity (ROE)

i. It is the amount of net income returned as a percentage of shareholders equity.
ii. Computation - Return on Equity (Net Income) / Average Stockholder Equity.

4. Debt To Equity Ratio

i. It measures the relationship between the borrowed capital (i.e. debt) and the capital contributed by shareholders (i.e. equity).
ii. Computation - Total Liabilities / Total Shareholder Equity.
iii. As a thumb of rule, companies with a debt-to equity ratio more than 1 are riskier.

Financial Ratios

5. Current Ratio

i. It is a key financial ratio for evaluating a company's liquidity.
ii. Computation - Current Assets / Current Liabilities.
iii. As a thumb rule, a company with a current ratio greater than 1 is better.

Technical Analysis

It focuses on the stock market, rather the company.

It seeks to predict price movements by examining historical data, mainly price and volume.

The underlying idea is that the market price already reflects the fundamentals of any given stock, which therefore can be ignored.

It is a good idea for investors to leverage both technical and fundamental analysis to fill the gaps.



Basic Terms Used in Study of Price Charts

Term	Meaning
Support Level	A level below which the price will likely not fall
Resistance Level	A level above which the price will not likely rise
Breakout	When a stock rises above its resistance level or falls below its support level
Trend line	A regression line that predicts future prices based on past prices
Relative strength	Ratio of the percentage price change of a stock to the percentage price change of a broader index or another stock

Difference between Fundamental & Technical Analysis

Fundamental analysis	Technical analysis
Analyses stock value based on economic and financial factors	Analyses historical stock movement to predict the future price of a stock
Long-term approach	Short-term approach
Uses financial statements for analysis	Uses price movement charts for analysis
Incorporates new market information	Focuses mainly on past performance

5 Steps Approach For Great Stock Picking

1. Approach stock purchases as buying a business rather than just a stock purchase in the portfolio.
2. Evaluate the true worth of the business considering the future earning potential.
3. The margin of safety is the real risk containment measure, and not stop loss.
4. Do not depend on turnaround as it seldom occurs.
5. Invest for the long term to generate inflation-adjusted superior returns.



So when one asks how to choose stock to invest in, the real question, is how to identify a great business and what all parameters should be used to identify it?

Price discounts everything

- "Each price represents a momentary consensus of value of all market participants – large commercial interests and small speculators, fundamental researchers, technicians and gamblers- at the moment of transaction" – Dr Alexander Elder
- Technical analysts believe that the current price fully reflects all the possible material information which could affect the price.

- The market price reflects the sum knowledge of all participants, including traders, investors, portfolio managers, buy-side analysts, sell-side analysts, market strategist, technical analysts, fundamental analysts and many others.
- It would be folly to disagree with the price set by such an impressive array of people with impeccable credentials.
- Technical analysis looks at the price and what it has done in the past and assumes it will perform similarly in future under similar circumstances.
- Technical analysis looks at the price and assumes that it will perform in the same way as done in the past under similar circumstances in future.

Price movements are not totally random

- Technical analysis is a trend following system.
- Most technicians acknowledge that hundreds of years of price charts have shown us one basic truth – prices move in trends.
- If prices were always random, it would be extremely difficult to make money using technical analysis.
- A technician believes that it is possible to identify a trend, invest or trade based on the trend and make money as the trend unfolds.
- Because technical analysis can be applied to many different time frames, it is possible to spot both short-term and long-term trends.

“What” is more important than “Why”

- It is said that “A technical analyst knows the price of everything, but the value of nothing”.
- Technical analysts are mainly concerned with two things:

1. The current price
2. The history of the price movement

- All of you will agree that the value of any asset is only what someone is willing to pay for it. Who needs to know why? By focusing just on price and nothing else, technical analysis represents a direct approach.
- The price is the final result of the fight between the forces of supply and demand for any tradable instrument.
- The objective of analysis is to forecast the direction of the future price.
- Fundamentalists are concerned with why the price is what it is. For technicians, the why portion of the equation is too broad and many times the fundamental reasons given are highly suspect.

- Technicians believe it is best to concentrate on what and never mind why. Why did the price go up? It is simple, more buyers (demand) than sellers (supply).
- The principles of technical analysis are universally applicable. The principles of support, resistance, trend, trading range and other aspects can be applied to any chart.
- Technical analysis can be used for any time horizon; for any marketable instrument like stocks, futures and commodities, fixed-income securities, forex, etc

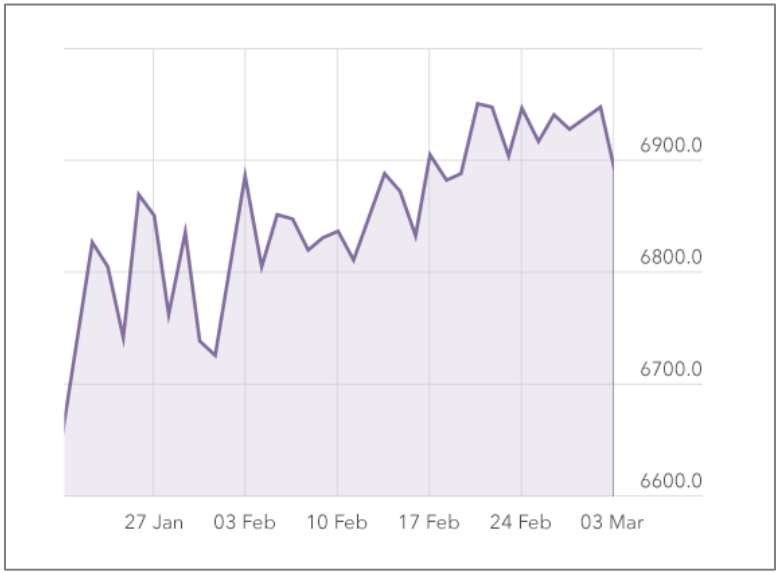
Price Charts

Technical analysts use a variety of charts based on the information they seek. However, there are three types of charts that are most commonly used. They are: Line, Bar and Candlestick

Line Chart

1. It plots the closing price of a share for each trading day over a period.

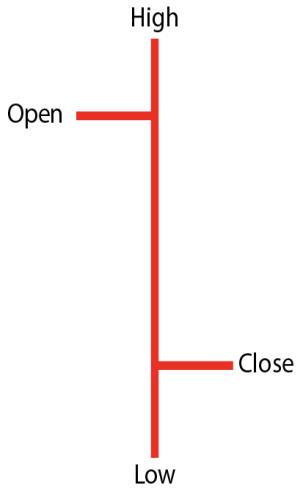
2. The line formed by joining the dots plotted on the graph shows the movements in stock price during the period.

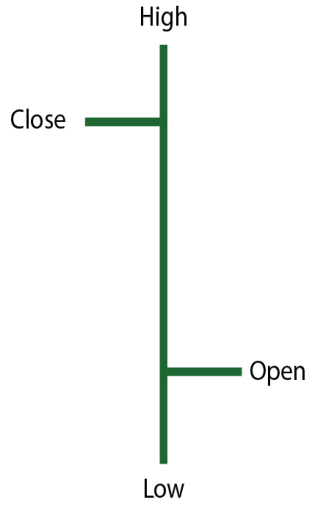


Price Charts

Bar Chart

1. It plots the intra-day high and low prices of a stock using a bar for each trading day for a specified time period.
2. The top of the bar corresponds to the day's high and the bottom, day's low.
3. Two additional horizontal lines indicate the opening and closing price. The length of the bar is proportional to the volatility in a stock.
4. Colored coded - If the share price closes above the open price it is colored green, and if the close is below the open the bar is colored red.

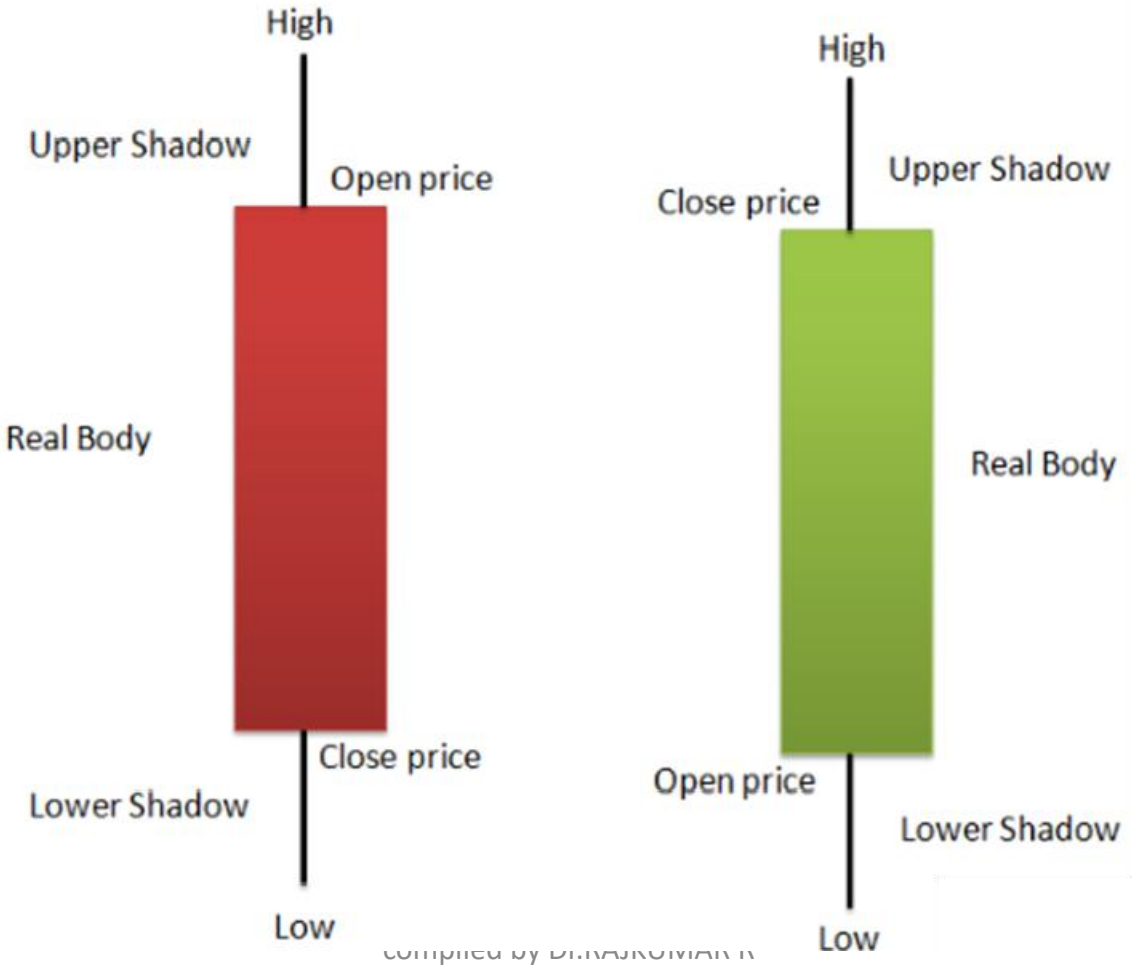




Price Charts

Candlestick chart

1. It displays the relationship between the high & low and opening & closing prices of a stock.
2. The body of the candle represents the opening and closing prices during the period.
3. Above and below the body are vertical lines called wicks or shadows that show the lows and highs of the traded prices.
4. While an individual candle provides sufficient information, patterns can be determined only by comparing one candle with its preceding and next candles.



compiled by Dr.RAJKUMAR R

CANDLE STICK CHARTS

- Candlestick charts originated in Japan over 100 years before the West had developed bar charts and point-and-figure charts.
- In the 1700s, a Japanese man known as Homma discovered that as there was a link between price and the supply and demand of rice, the markets were also strongly influenced by the emotions of traders.
- A daily candlestick chart shows the security's open, high, low, and close prices for the day. The candlestick's wide or rectangle part is called the "real body" which shows the link between opening and closing prices.

- This real body shows the price range between the open and close of that day's trading.

- When the real body is filled, black or red, then it means that the close is lower than the open and is known as the bearish candle. It shows that the prices opened, the bears pushed the prices down and closed lower than the opening price.

- If the real body is empty, white, or green, then it means that the close was higher than the open, known as the bullish candle. It shows that the prices opened, the bulls pushed the prices up and closed higher than the opening price.
- The thin vertical lines above and below the real body are known as the wicks or shadows, which represent the high and low prices of the trading session.
- The upper shadow shows the high price, and lower shadow shows the low prices reached during the trading session.

ASSUMPTIONS

- there are a few assumptions that need to be kept in mind that are specific to the candlestick charts.

1. Strength is represented by a bullish or green candle, and weakness by a bearish or red candle. One should ensure that whenever they are buying, it is a green candle day, and whenever they are selling, ensure that it's a red candle day.
2. The textbook definition of a pattern states certain criteria, but one should state that there could be minor variations to the pattern depending on certain market conditions.
3. One should look for a prior trend. If you are looking at a bullish reversal pattern, then the prior trend should be bearish, and if you are looking for a bearish reversal pattern, then the prior trend should be bullish.

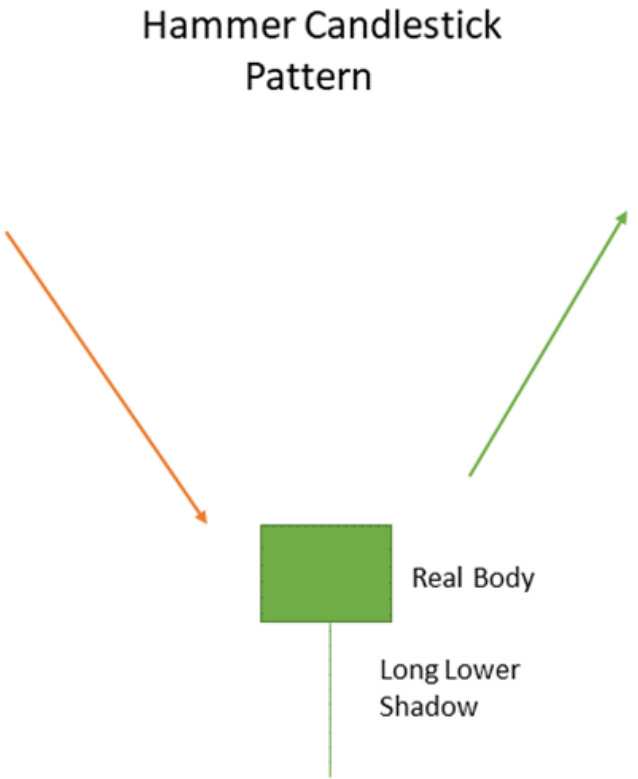
CANDLE STICK PATTERNS

1.One candle pattern
2. Two candle pattern
3.Three candle pattern

One candle pattern

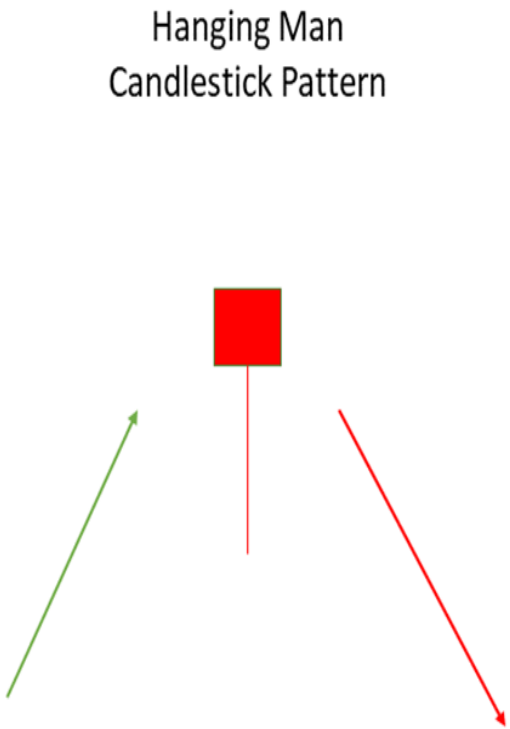
Hammer

- Hammer is a single candlestick pattern that is formed at the end of a downtrend and signals a bullish reversal.
- The real body of this candle is small and is located at the top with a lower shadow which should be more than twice the real body. This candlestick chart pattern has no or little upper shadow.
- The psychology behind this candle formation is that the prices opened, and sellers pushed down the prices.
- Suddenly the buyers came into the market and pushed the prices up and closed the trading session more than the opening price.



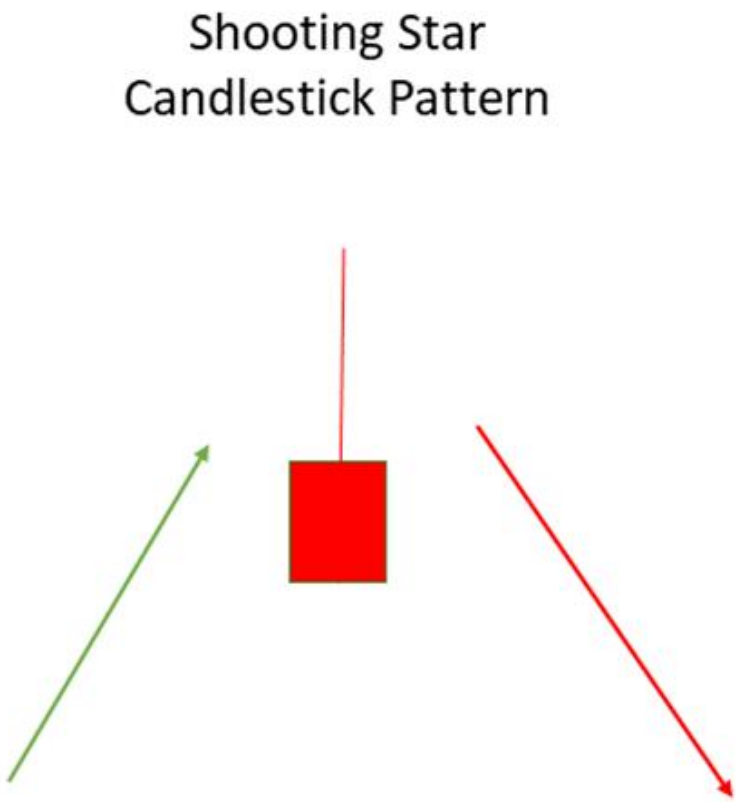
Hanging man

- Hanging Man is a single candlestick pattern that is formed at the end of an uptrend and signals a bearish reversal.
- The real body of this candle is small and is located at the top with a lower shadow which should be more than twice the real body. This candlestick pattern has no or little upper shadow.
- The psychology behind this candle formation is that the prices opened, and the seller pushed down the prices.
- Suddenly the buyers came into the market and pushed the prices up but were unsuccessful in doing so, as the prices closed below the opening price.



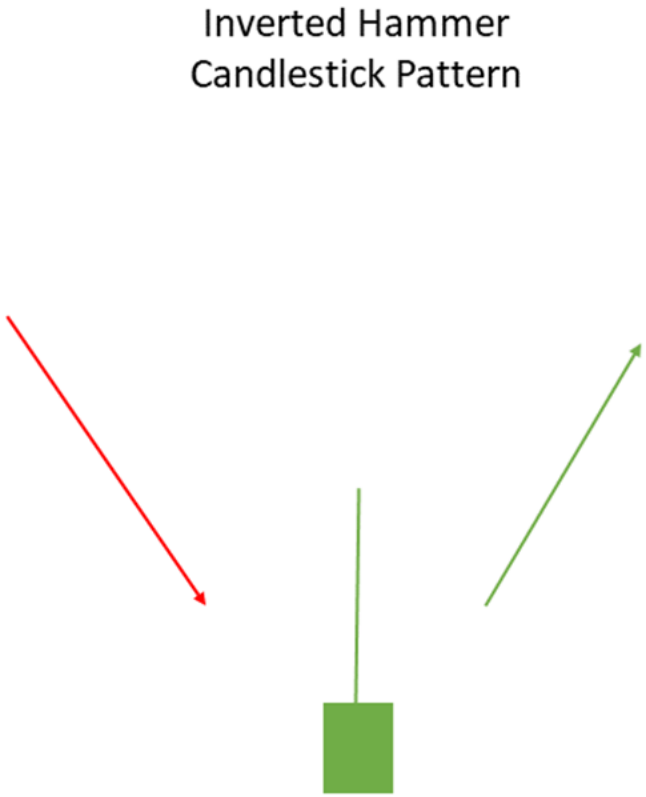
Shooting star

- Shooting Star is formed t the end of the uptrend and gives a bearish reversal signal.
- In this candlestick chart, the real body is located at the end, and there is a long upper shadow. It is the inverse of the Hanging Man Candlestick pattern.



Inverted hammer

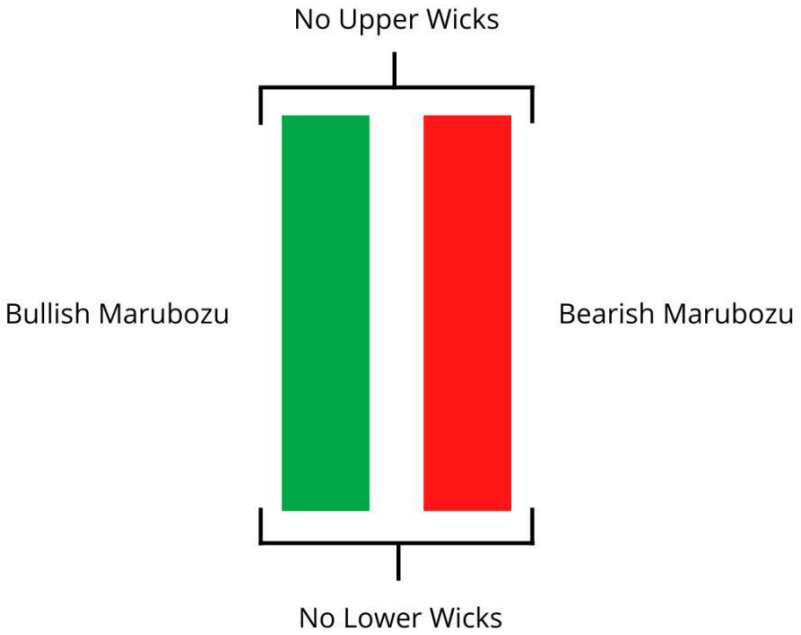
- An Inverted Hammer is formed at the end of the downtrend and gives a bullish reversal signal.
- In this candlestick, the real body is located at the end, and there is a long upper shadow. It is the inverse of the Hammer Candlestick pattern.
- This pattern is formed when the opening and closing prices are near to each other, and the upper shadow should be more than twice the real body



ONE CANDLE PATTERN

Marubozu

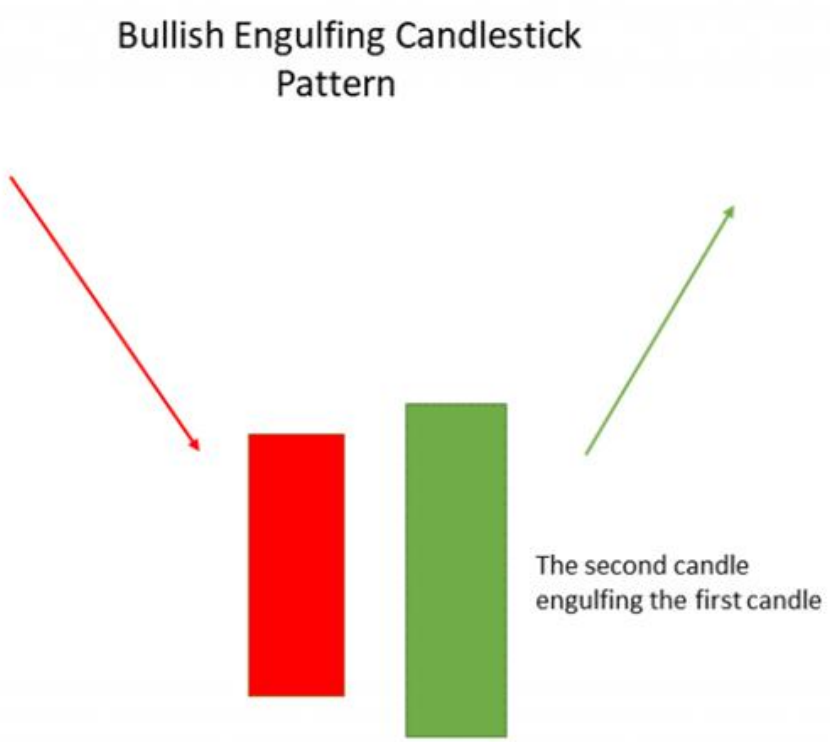
- A long-bodied candle with very small or no shadows in its textbook form.
- Interpretation: strong directional movement during the period.



Two candle pattern

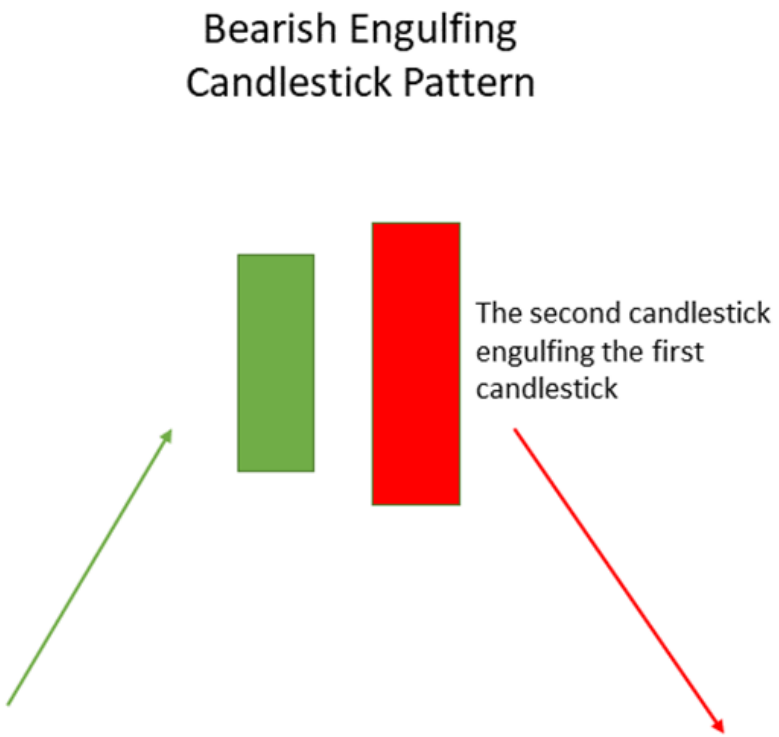
Bullish engulfing

- Bullish Engulfing is a multiple candlestick chart pattern that is formed after a downtrend indicating a bullish reversal.
- It is formed by two candles, the second candlestick engulfing the first candlestick. The first candle is a bearish candle that indicates the continuation of the downtrend.
- The second candlestick is a long bullish candle that completely engulfs the first candle and shows that the bulls are back in the market.



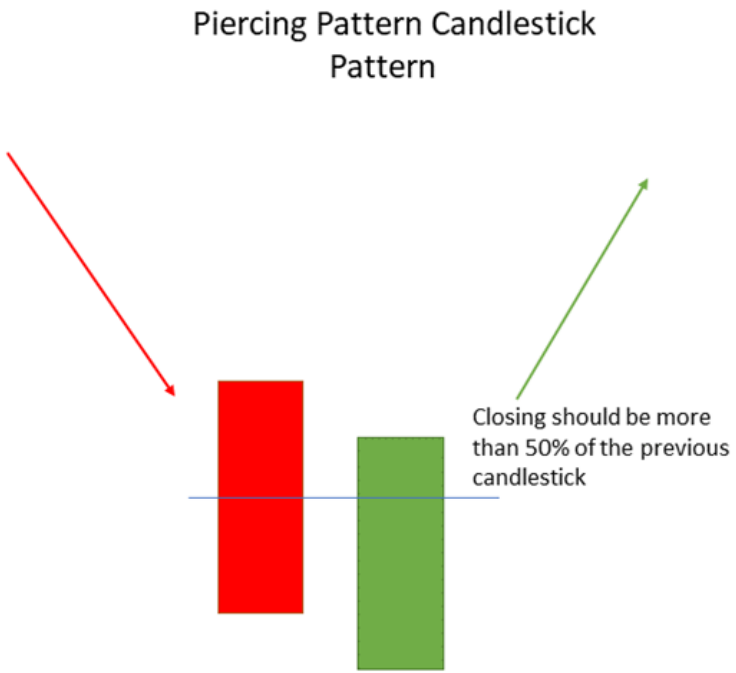
Bearish engulfing

- Bearish Engulfing is a multiple candlestick pattern that is formed after an uptrend indicating a bearish reversal.
- It is formed by two candles, the second candlestick engulfing the first candlestick. The first candle being a bullish candle indicates the continuation of the uptrend.
- The second candlestick chart is a long bearish candle that completely engulfs the first candle and shows that the bears are back in the market.



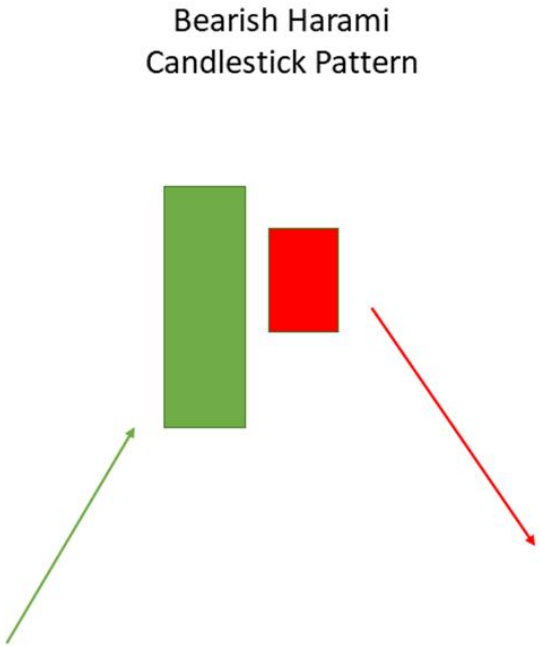
Piercing

- Piercing pattern is a multiple candlestick chart pattern formed after a downtrend indicating a bullish reversal.
- Two candles form it, the first candle being a bearish candle which indicates the continuation of the downtrend.
- The second candle is a bullish candle which opens the gap down but closes more than 50% of the real body of the previous candle, which shows that the bulls are back in the market and a bullish reversal is going to take place.



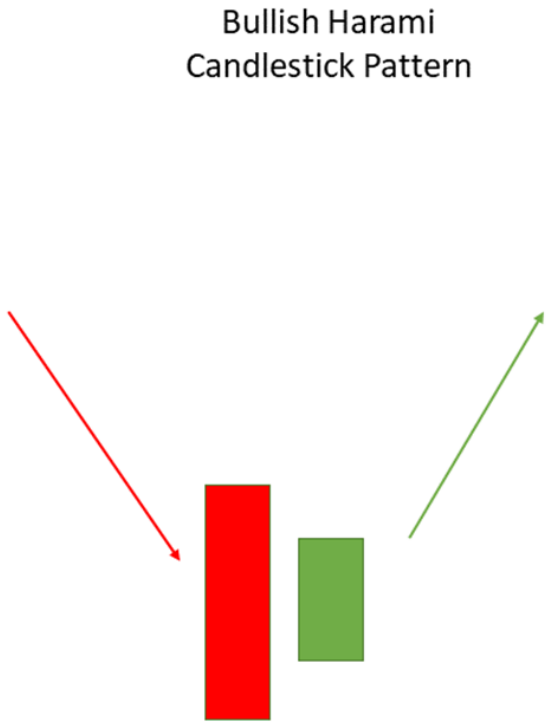
Bearish harami

- The Bearish Harami is a multiple candlestick pattern formed after the uptrend indicating bearish reversal.
- It consists of two candlesticks, the first candlestick being a tall bullish candle and second being a small bearish candle which should be in the range of the first candlestick chart.
- The first bullish candle shows the continuation of the bullish trend and the second candle shows that the bears are back in the market.

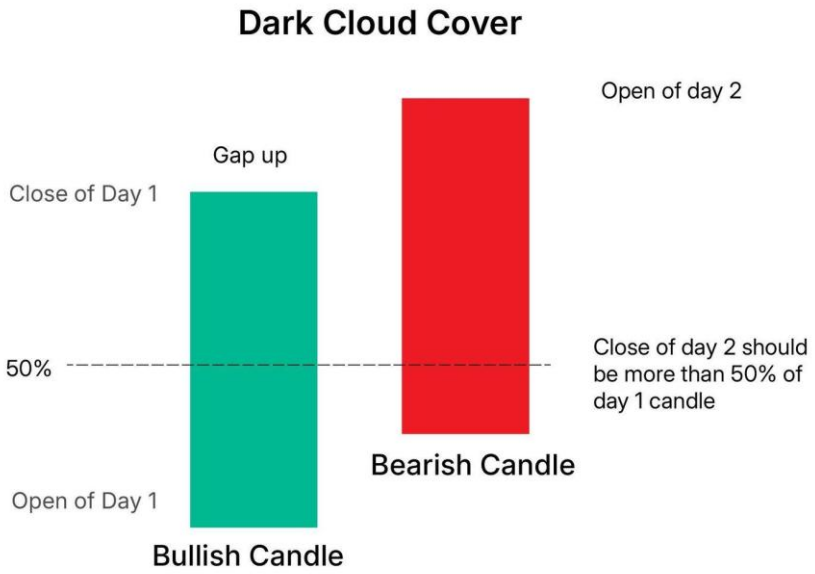


Bullish harami

- The **Bullish Harami** is multiple candlestick chart pattern which is formed after a downtrend indicating bullish reversal.
- It consists of two candlestick charts, the first candlestick being a tall bearish candle and second being a small bullish candle which should be in the range of the first candlestick.
- The first bearish candle shows the continuation of the bearish trend and the second candle shows that the bulls are back in the market.



Dark Cloud Cover



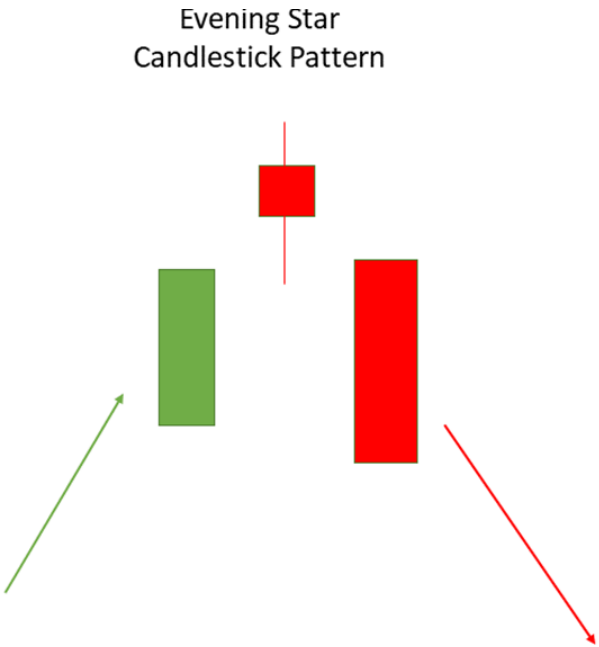
- Structure: Following a bullish candle, a bearish candle opens above the previous close and closes below the midpoint of the previous real body, without closing below its open under the conventional definition.
- Interpretation: The second candle reverses a substantial portion of the prior gain, potentially signalling weakening buying pressure.

Three candle pattern

1. Evening star
2. Morning star
3. Doji

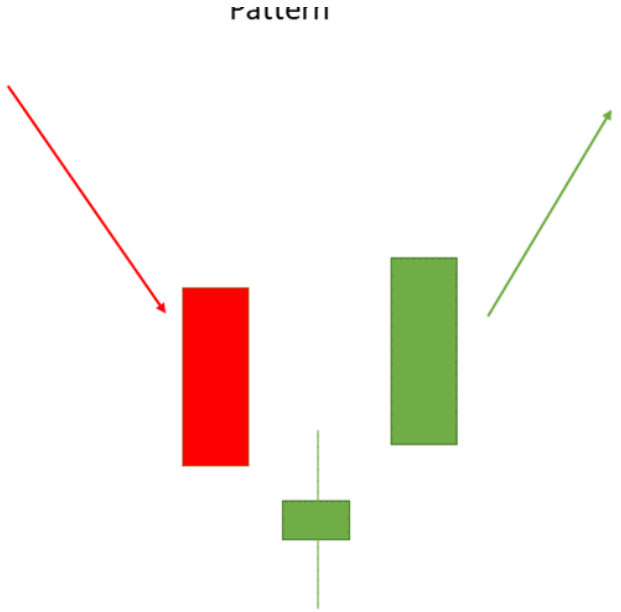
The Evening Star

- The Evening Star is a multiple candlestick pattern that is formed after the uptrend indicating a bearish reversal.
- It is made of 3 candlesticks, the first being a bullish candle, the second a doji, and the third being a bearish candle.
- The first candle shows the continuation of the uptrend, the second candle being a doji indicates indecision in the market, and the third bearish candle shows that the bears are back in the market and reversal is going to take place.
- The second candle should be completely out of the real bodies of the first and third candles.
- Traders can enter a long position if next day a bearish candle is formed and can place a stop-loss at the high of the second candle.



The Morning Star

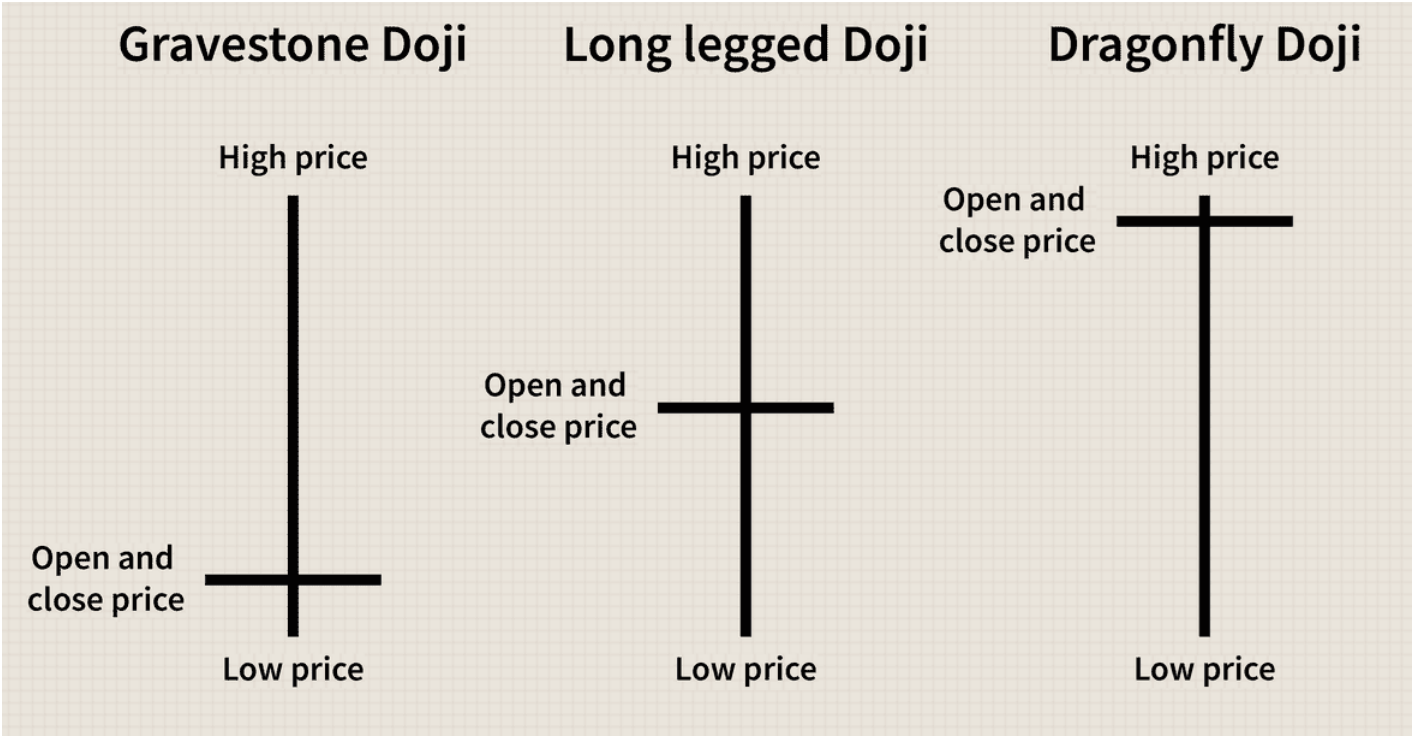
- The Morning Star is a multiple candlestick chart pattern that is formed after a downtrend indicating a bullish reversal.
- It is made of 3 candlesticks, the first being a bearish candle, the second a Doji and the third being a bullish candle.
- The first candle shows the continuation of the downtrend. The second candle being a doji indicates indecision in the market. The third bullish candle shows that the bulls are back in the market and reversal will take place.
- The second candle should be completely out of the real bodies of the first and third candles.
- Traders can enter a long position if the next day a bullish candle is formed and can place a stop-loss at the low of the second candle.





Variations in Doji

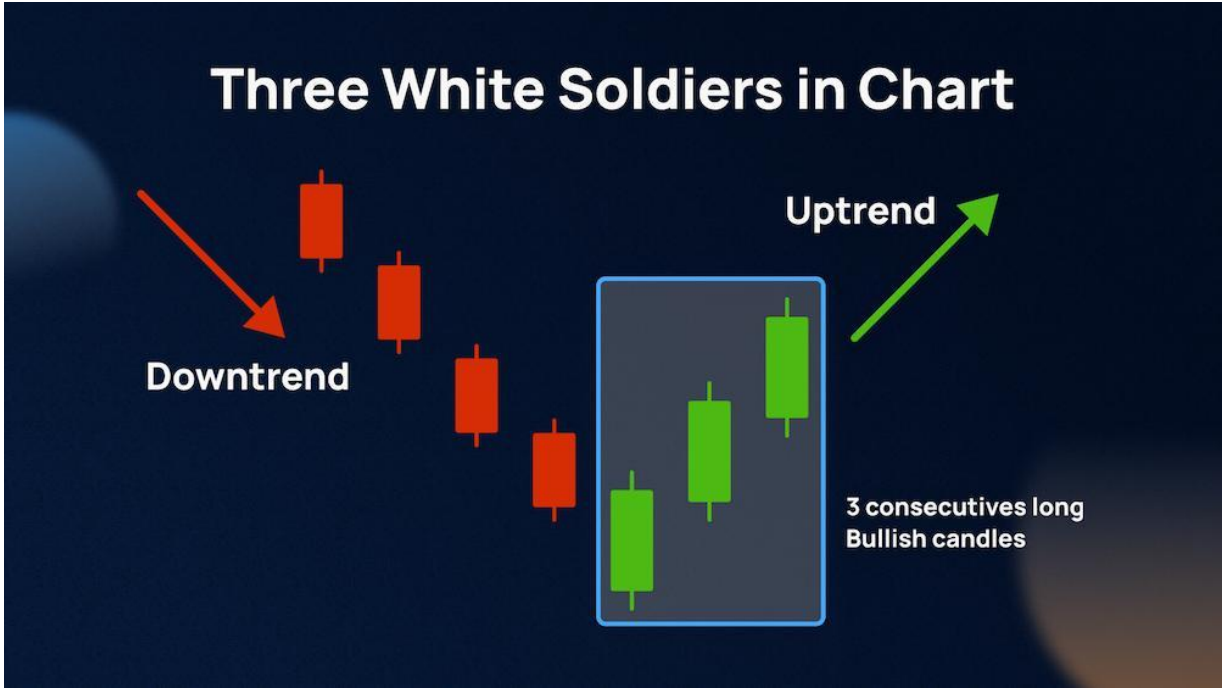
- When the Doji does not have a lower shadow, it is termed a **Gravestone Doji**.
- The Japanese compare it to the soldiers going out to battle during the day and coming back to their original positions for rest during the night.
- This type of Doji formation is very effective at the top of a trend. The logic being that though the bulls tried to rally the stock they failed at the end of the day.
- A Doji without any upper shadow is known as a **Dragonfly Doji**, just because it looks like one. This kind of Doji would have powerful implications near the bottom of the trend.
- It implies that while the bears tried to take the price down during the day, by the end the bulls were able to bring the price back to where it had opened.When the Doji has a small body, it is also known as a **Spinning Top**


Gravestone Doji
High price
Open and close price
Low price
Long legged Doji
High price
Open and close price
Low price
Dragonfly Doji
High price
Open and close price
Low price

Spinning Top
Candlestick Pattern

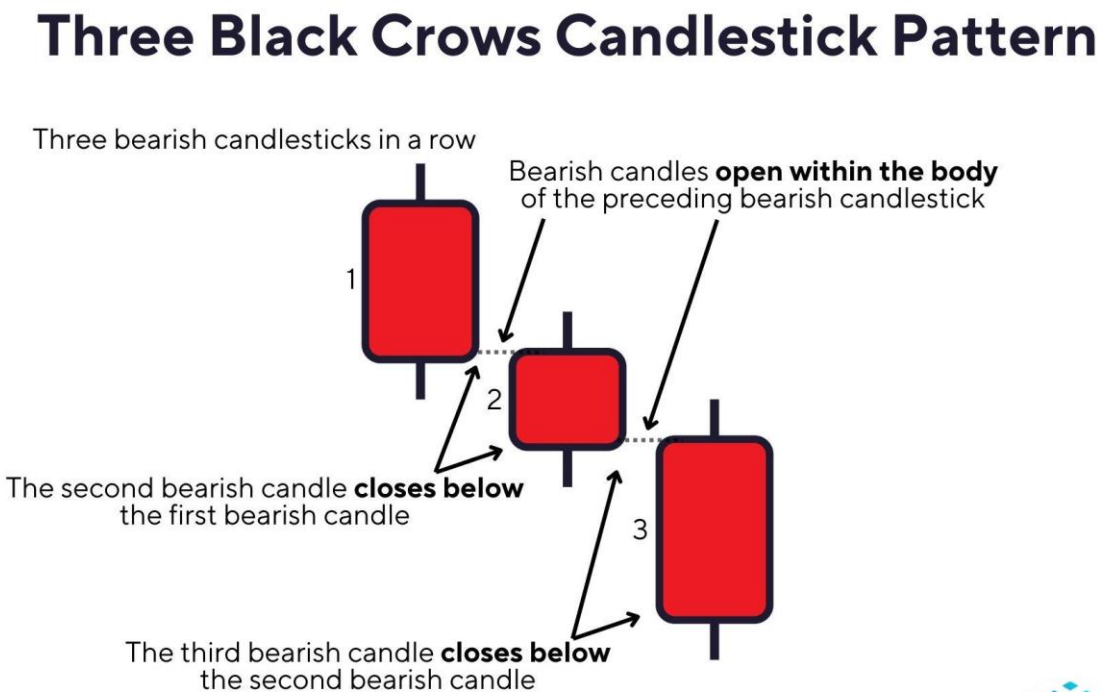


Three White Soldiers



- Structure: Three consecutive bullish candles with relatively strong bodies and progressively higher closes.
- Interpretation: Sustained buying pressure may be developing.

Three Black Crows



- Structure: Three consecutive bearish candles with relatively strong bodies and progressively lower closes.
- Interpretation: Sustained selling pressure may be developing.
- Discuss why a sequence of bearish candles near an established support zone may require a different response from the same sequence after support has already broken.

The Essential Trading Principle

PATTERN + LOCATION + CONFIRMATION

- 1. Pattern
- Is the candle combination correctly identified according to the chosen definition?
- 2. Location
- Does it occur after a decline, after an advance, near support/resistance, or within a range?
- 3. Confirmation
- What happens next? Does price follow through, contradict the interpretation, or remain inconclusive?

SUPPORT AND RESISTANCE

- Remember, **candlesticks are useless on their own, and you must always consider the market environment and what the price is telling you**.
- **As with any technical indicator or tool, if candlesticks point to a reversal or continuation that does NOT mean it will happen.**
- **This is the forex market, and nothing is set in stone!**
- Using Candlesticks with Support and Resistance
- Because support and resistance levels determine areas, where buyers and sellers have set up their defenses, looking at how candlesticks react to them, will help you greatly in predicting where price will head next.

INTRODUCTION

- Support and resistance are the important price points on the chart of any security at which the security is expected to attract the maximum amount of buying or selling
- They are the points where there is a maximum probability of a pause or a reversal of a prevailing trend
- The logic behind them is demand and supply
- The price of any financial asset or security is driven due to the law of demand and supply
- When the demand (buyers) is more than the supply(sellers) price of security increases
- When the supply (sellers) is more than the demand(buyers) price of security decreases

- The age old saying ‘Buy low and sell high’ – is the only thing is required to make money in any financial market (stocks,commodities,currencies,crypto currencies etc.)
- For this , one must either buy at support and later sell at resistance
- Sell at resistance and buy back later at support

What is support?

- It is the level at which the price of a security does not fall further as the demand is sufficient to hold the price from falling further
- As price reaches support level , more and more buyers get keen to buy the security whereby demand becomes more than supply and hence price stops from falling further and starts rising
- **Support will always appear or form below the current market price of a security**
- Support can either be in form of a single price level or a zone
- Support level or zone ultimately gives us the reference point where the most repetitive buying (demand) has happened in the past and is likely to happen again in the future

How to plot support?

- 1.look at the historical price data
- 2.identify the extreme lows
- 3.Mark all the low touch points
- 4.Connect all the touch point

What is resistance?

- It is the level at which the price of a security does not rise further as the supply is sufficient enough to hold the price from rising further
- When demand is more than supply, the price of a security increases
- When the price reaches the resistance level, more and more sellers get keen to sell the security whereby supply becomes more than demand and hence price stops from rising further and starts falling
- Resistance will always appear or form above the current market price of a security
- Resistance can either be in form of a single price level or a zone
- Support can either be in form of a single price level or a zone
- Resistance level or zone ultimately gives us the reference point where the most repetitive selling (supply) has happened in the past and is likely to happen again in the future

How to plot resistance?

- 1.look at the historical price data
- 2.identify extreme highs
- 3.mark all high touch points
- 4.connect all the touch points